

Case Study

Project Balance — ₹100 Crore Cash Release Without Hurting Service

You have joined NovaTech Manufacturing Ltd. as a Supply Chain Analytics Manager. NovaTech manufactures industrial equipment and components and serves customers across India through five regional warehouses. The company has been growing, but its supply-chain performance is creating tension between Finance and Sales.

Current Situation

KPI	Current
Annual Revenue	₹2,400 Cr
Annual COGS	₹1,800 Cr
Inventory	₹500 Cr
OTIF	92%
Stockout Rate	7.5%
Forecast Accuracy	76%
Supplier OTD	88%
Average Lead Time	24 days
Inventory Turns	3.6×
Days Inventory	~101 days

Challenge

The CFO says: “₹500 crore is too much inventory. Release at least ₹100 crore.”

Therefore: Inventory target = ₹500 Cr → ₹400 Cr

At the same time, the Sales Head says:

“Customer complaints are increasing. OTIF must exceed 97%.”

The CEO turns to you:

“Can we reduce inventory by 20% while increasing OTIF from 92% to above 97%? What would you do?”

Write down your approach and then scroll to next page

HINT

Attempt your answer before proceeding ahead...

Before recommending an across-the-board inventory reduction, I would determine whether the ₹500 crore is the right inventory in the wrong products or locations. I would segment the portfolio, diagnose the drivers of excess inventory and poor OTIF, model differentiated interventions, and then determine whether ₹100 crore can be released without compromising service.

Approach:

SEGMENT → DIAGNOSE → MODEL → DECIDE → MEASURE

SEGMENT	DIAGNOSE	MODEL	DECIDE	MEASURE
• Where should we look?	• What's causing the problem?	• What are our options and trade-offs?	• What should we do?	• Did it create business value?

SOLUTION APPROACH

STEP 1 — SEGMENT

You ask for more information. The analytics team gives you the following.

Segment	% SKUs	Inventory	OTIF	Demand Characteristic
AX	10%	₹110 Cr	98%	High-value, stable
AY	12%	₹90 Cr	94%	High-value, variable
AZ	8%	₹65 Cr	87%	High-value, unpredictable
BX/BY	25%	₹100 Cr	94%	Medium-value
BZ	15%	₹60 Cr	90%	Medium-value, unpredictable
CX/CY/CZ	30%	₹75 Cr	96%	Low-value
Total	100%	₹500 Cr	92%	

Would you reduce every segment by 20%? No.

AX inventory may be highly productive. Some CZ inventory may have little commercial value despite occupying working capital. AZ may actually require better availability, not indiscriminate reduction. That establishes differentiated inventory policy.

STEP 2 — DIAGNOSE

Where is the ₹500 crore?

Analytics discovers:

Inventory Condition	Value
Healthy/productive inventory	₹300 Cr
Excess safety stock	₹55 Cr
Slow-moving inventory	₹48 Cr
Obsolete inventory	₹27 Cr
Excess caused by MOQ	₹20 Cr
Inventory in wrong warehouse	₹30 Cr
Customer/project-specific excess	₹20 Cr
Total	₹500 Cr

Now the case becomes interesting. There is potentially: ₹200 crore of inventory requiring management attention.

The CFO only wants ₹100 crore released. Therefore, the answer isn't necessarily: Reduce everything.

It could be: *Remove nonproductive inventory while protecting or even increasing inventory where service requires it.*

STEP 3 — Why is OTIF Only 92%?

Give another diagnostic table.

OTIF Failure Driver	Contribution to Failed Orders
Supplier delays	30%
Inventory in wrong location	22%
Forecast error/bias	18%
Production/capacity constraints	12%
Transportation delays	8%
Order/data issues	6%
Other	4%

“Does the evidence suggest that the primary solution to poor OTIF is simply more inventory?” No.
That's the key insight. A large portion of service failure comes from supplier reliability, positioning, forecasting and operational execution.

STEP 4 — Find the Geographic Problem

Region	Inventory	OTIF
North	₹80 Cr	84%
South	₹115 Cr	97%
East	₹80 Cr	93%
West	₹145 Cr	96%
Central	₹80 Cr	94%

North • High demand • Low inventory availability • Long replenishment lead time	West • Moderate demand • High inventory • Several slow-moving SKUs
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“What would you do?”

Possible Solution: Rebalance before buying. Move appropriate inventory: West → North

Instead of: Supplier → North

This simultaneously improves service and avoids additional procurement.

STEP 5 — Forecast Diagnosis

“Averages hide the problem.”

Company Forecast Accuracy = 76%

Segment	Forecast Accuracy	Bias
AX	91%	+1%
AY	80%	-4%
AZ	61%	-14%
BX/BY	79%	-2%
BZ	65%	-8%
CX/CY/CZ	83%	+11%

Observations:

- **AZ:** Underforecasting → shortage → OTIF problem.
- **C-items:** Overforecasting → excess inventory.

Analytical insight: *The same forecasting process is simultaneously creating shortages in one segment and excess inventory in another.*

STEP 6 — Supplier Diagnosis

Supplier	Spend	OTD	Lead Time	Critical SKUs
Supplier A	₹300 Cr	97%	15 days	22
Supplier B	₹220 Cr	94%	18 days	18
Supplier C	₹180 Cr	71%	36 days	31
Supplier D	₹120 Cr	91%	22 days	12
Others	₹280 Cr	93%	20 days	27

“What would happen if we simply increase safety stock for Supplier C items?”

Possible Solution: It may improve availability—but at the cost of additional working capital. A better strategic response could include: supplier development + dual sourcing + lead-time reduction + targeted buffer; rather than permanently solving supplier unreliability with inventory.

STEP 7 — MODEL SCENARIOS

Scenario	Inventory	Reduction	Expected OTIF	Additional Annual Cost
Current	₹500 Cr	—	92.0%	—
A: Across-the-board –20%	₹400 Cr	₹100 Cr	86.5%	₹5 Cr
B: Reduce slow/excess inventory	₹410 Cr	₹90 Cr	93.5%	₹8 Cr
C: Service-first, add buffers	₹525 Cr	–₹25 Cr	97.8%	₹18 Cr
D: Segmented optimization	₹398 Cr	₹102 Cr	97.3%	₹14 Cr

Scenario A

- Meets CFO's inventory objective.
- Fails Sales objective.

Scenario C

- Meets Sales objective.
- Fails CFO objective.

Scenario D

- Potentially satisfies both.

This introduces multi-objective optimization without requiring students to understand optimization algorithms.

STEP 8 — What Does Scenario D Actually Do?

₹500 Cr current inventory.

Then identify releases.

1. Dispose/liquidate obsolete inventory

- ₹27 Cr → release ₹22 Cr
- Some residual/recovery inventory remains.

2. Reduce slow-moving inventory

- ₹48 Cr → release ₹25 Cr

3. Correct excessive safety stock

- ₹55 Cr → release ₹18 Cr
- But only from stable/reliable segments.

4. Renegotiate MOQ / order frequency

- Release ₹12 Cr

5. Rebalance warehouses

- Release duplicate regional buffers worth ₹15 Cr

6. Reduce project/customer-specific excess

- Release ₹10 Cr

Gross release: ₹102 Cr

Inventory: ₹500 Cr – ₹102 Cr = ₹398 Cr

So Finance's requirement is achieved.

But that's only half the problem.

STEP 9 — How Do We Raise OTIF?

1. Fix Supplier C

OTD: 71% → target >93%

Actions:

- dual source selected critical SKUs
- supplier capacity review
- supplier SLA
- weekly exception monitoring

Expected OTIF contribution: +1.6 percentage points

2. Rebalance inventory

Move inventory from: West → North, rather than purchasing additional stock.

Expected OTIF: +1.1 points

3. Correct forecast bias

Focus particularly on AZ and BZ.

Expected improvement: +0.8 points

4. Differentiate safety stock

- Don't reduce all safety stocks.
- AX: Lower buffers possible because demand is stable.
- AZ: Potentially maintain/increase targeted buffer.
- CZ: Minimize stock / consider order-to-demand.

Expected: +0.6 points

5. Logistics and order-priority rules

Prioritize:

- critical customers
- high-margin orders
- contractual SLA customers

Expected: +0.5 points

6. Exception-based control tower

Alert planners before:

- stockout
- supplier delay
- capacity shortage
- forecast deviation

Expected: +0.7 points

Combined modeled OTIF result: 92.0% → 97.3%

STEP 10 — The Recommendation

Presentation to the CEO (Don't give a 10-minute technical explanation.)

A strong executive answer would sound like this:

I would not reduce inventory uniformly by 20%, because that could worsen our existing service problem. My analysis indicates that the issue is not simply the quantity of inventory but its composition, positioning and replenishment reliability.

I would segment inventory by value and variability, remove obsolete and slow-moving stock, correct excessive safety stocks and MOQs, and rebalance inventory across regions. That can potentially release approximately ₹102 crore, reducing inventory from ₹500 crore to about ₹398 crore.

In parallel, I would address OTIF through targeted interventions: improve Supplier C reliability, correct forecast bias in critical SKUs, reposition inventory toward shortage locations, differentiate safety stocks and implement exception-based monitoring.

Our modeled scenario indicates OTIF could improve from 92% to approximately 97.3% while inventory falls by roughly 20%. I would pilot this on the highest-impact categories and regions before scaling.”

STEP 11 — Convert the Result Into Business Value

Metric	Before (Current State)	After (Improved Planning)	Change	Business Value / Impact
 Inventory Value (Working Capital) ₹ Crore	₹500 Cr	₹398 Cr	↓ ₹102 Cr released	₹102 CRORE cash released for growth, debt reduction or other strategic investments
 Inventory Turns Times per year	3.6x	4.5x	↑ +0.9x	Higher efficiency and better utilisation of inventory
 Days Inventory Days	101	~81	↓ ~20 days	Lower holding cost and reduced working capital requirement
 OTIF (On-Time In-Full)	92%	97.3%	↑ +5.3 pp	Higher customer satisfaction, stronger relationships and more reliable service performance

Potential Secondary Effects
 Improved planning and forecast accuracy can also lead to:


Fewer lost sales
 Capture more demand and revenue


Lower expediting
 Reduced premium freight costs


Lower obsolescence
 Less waste and markdowns


Less warehouse space pressure
 Delay or avoid expansion costs


Better customer experience
 Higher loyalty and repeat business

“The recommendation releases approximately ₹102 crore of working capital while improving modeled OTIF by 5.3 percentage points.”

STEP 12 — Implementation Roadmap

0 – 30 DAYS STOP THE BLEEDING	30 – 90 DAYS OPTIMIZE	3 – 6 MONTHS PREDICT	6 – 12 MONTHS ORCHESTRATE
Stabilize operations and address immediate risks.  Identify stockout-critical SKUs Focus on high-risk items  Freeze unnecessary purchases of obsolete/slow movers Preserve cash and space  Rebalance regional inventory Move stock to where it's needed  Create Supplier C recovery plan Mitigate supply risk  Introduce daily OTIF exceptions Track and act on service issues	Improve efficiency and build a stronger foundation.  ABC-XYZ segmentation Classify SKUs by value and variability  Recalculate safety stocks Right-size buffers  Correct forecast bias Address systematic over/under forecasting  Renegotiate MOQs Improve flexibility and cash flow  Dual-source critical items Reduce single-source risk  Implement inventory policies by segment MTS, ATO, MTO based on product profile	Use data and analytics to see what's coming.  Predict stockouts Identify risk early  Predict supplier delays Anticipate disruptions  Demand sensing Use real-time and external signals  Inventory optimization models Optimize stock across network  Control-tower alerts Proactive, exception-based monitoring	Move from insights to autonomous, end-to-end execution.  Prescriptive recommendations Best action with business constraints  Scenario simulation Test multiple what-if scenarios  Digital twin Virtual model of the supply chain  AI-assisted planning Human + AI collaboration  Agentic workflow automation Execute approved actions across systems

STEP 13 — Management Control Tower

The CEO scorecard should balance:

KPI	Baseline	Target
Inventory	₹500 Cr	≤₹400 Cr
OTIF	92%	≥97%
Inventory Turns	3.6×	≥4.5×
Days Inventory	101	≤82
Stockout Rate	7.5%	<3%
Forecast Accuracy	76%	>85%
Supplier OTD	88%	>95%
Expedite Cost	Baseline	−20%

This reinforces another important lesson: Never optimize one KPI in isolation.

If inventory drops but OTIF collapses, the program failed. If OTIF rises because inventory increased by ₹200 crore, the program may also have failed economically.

Summary

BUSINESS PROBLEM → DATA → INSIGHT → DECISION → VALUE

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